



Name: _____

Date: _____

SOLUTIONS

Year 9 Mathematics

Test 8, 2015

60
= %

Total Time: 64 minutes

Total Reading: 4 minutes

Total Working: 60 minutes

Weighting: % of the year

Equipment: Curriculum Council Formulae Sheet, 1/2 page notes (A4 one side), CAS calculator, Scientific Calculator

SECTION 1: CALCULATOR FREE

Time:	20 minutes	Marks for Section 1:	18
Reading:	2 minutes	Equipment Allowed:	Nil
Working:	18 minutes		

1. [3 marks: 1, 1, 1]

Circle the correct answer:

a) What is the correct general form for a linear equation:

i) $c = x - y$

ii) $y = mx + c$

iii) $x^2 = ym + c$

iv) $y = cm + x$

(D/E)

b) A line with a gradient of -2 and a y-intercept of -1 would have an equation of:

i) $y = x - 2$

ii) $y = -2x + 1$

iii) $y = -2x - 1$

iv) $y = -x - 2$

c) The graph of the rule $y = -2$, would be a _____ line?

i) sloping

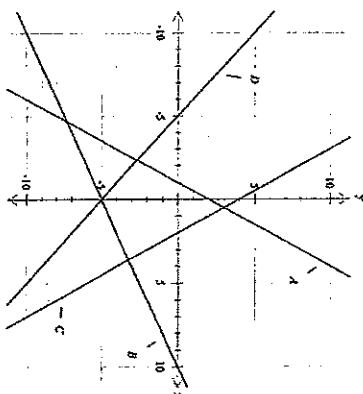
ii) vertical

iii) curved

iv) horizontal

2. [4 marks: 1, 1, 1, 1]

a) Match the each graph to one of the equations below:



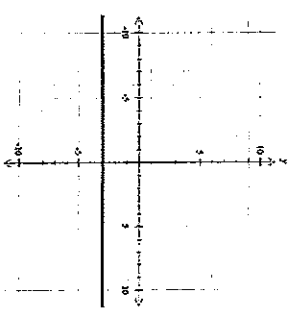
(C)

Equation	$y = -x - 5$	$y = \frac{1}{2}x - 5$	$y = 2x + 2$	$y = -2x + 4$
Graph	D	B	A	C

3. [2 marks: 1, 1]

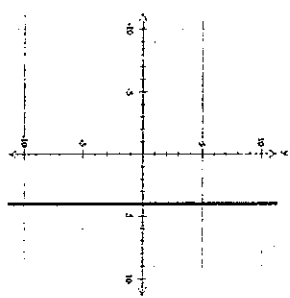
Write down the equation of each of these lines:

a)



$y = -3$

b)



$x = 4$

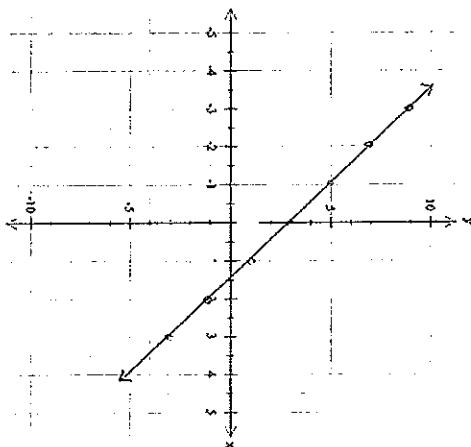
(C)

4. [3 marks: 1, 2]

Complete the table using the given rule, then plot on the Cartesian Plane:

$$y = -2x + 3$$

x	-3	-2	-1	0	1	2	3
y	9	7	5	3	1	-1	-3



5. [6 marks: 1, 1, 2, 2]

Solve the following linear equations

a) $x + 7 = 23$

(-7)

b) $\frac{x}{3} = 12$

$(\times 3)$

(D)

$x = 16$

$x = 36$

c) $2x - 2 = 20$

$(+2)$

d) $\frac{3x}{5} = 6$

$(\times 5)$

$2x = 22$

$(\div 2)$

$3x = 30$

$(\div 3)$

$x = 11$

$x = 10$

Name: _____

Solutions

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time:	44 minutes	Marks for Section 2:	42
Reading:	2 minutes	Equipment Allowed:	1/2 page notes (A4 one side), CAS calculator
Working:	42 minutes		

6. [12 marks: 2, 2, 2, 2, 2]

Solve the following linear equations:

a) $6x = -20 + x$

$(-x)$

$5x = -20$

$(\div 5)$

$x = -4$

$x + 6 = 45$

(-6)

$x = 39$

c) $\frac{x}{3} + \frac{1}{4} = \frac{1}{2}$

$(\cdot \frac{1}{4})$

$\frac{x}{3} = \frac{1}{4}$

$(\times 3)$

$x = \frac{3}{4}$

d) $5(x - 3) = 2x$

$5x - 15 = 2x$

$-15 = -3x$

$5 = x$

$(-5x)$

$(\div -3)$

e) $5(x - 12) = 2(x - 3)$

$5x - 60 = 2x - 6$

$(-2x)$

$3x - 60 = -6$

$(+60)$

$3x = 54$

$(\div 3)$

$x = 18$

f) $\frac{11 - 2x}{4} + 3 = -6$

(-3)

$\frac{11 - 2x}{4} = -9$

$(\times 4)$

$11 - 2x = -36$

(-11)

$-2x = -47$

$(\div -2)$

$x = 23.5$

7. [4 marks: 2, 2]

Write the rule that relates y to x for each of the following table of values:

a)

x	0	1	2	3	4
y	10	6	2	-2	-6

$$y = -4x + 10$$

(A)

b)

x	1	2	3	4	5
y	-7	-3	1	5	9

$$y = 4x - 11$$

8. [4 marks: 2, 2]

Identify the gradient and y-intercept for each of these equations:

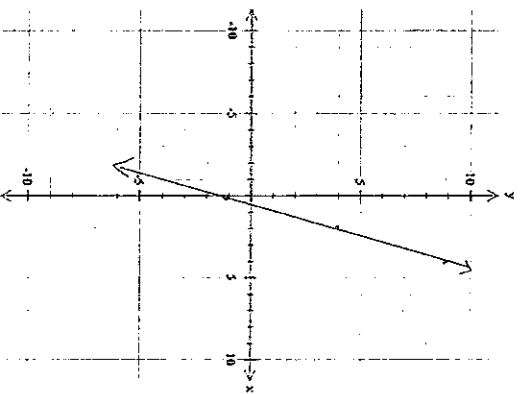
Equation	Gradient	y-intercept
$y = -2x - 3$	-2	-3
$2y = 3x + 6$	$\frac{3}{2}$	3

(C)

9. [3 Marks]

Graph the line for the equation: $2y = 5x - 2$, using any suitable method.

(B)



$$y = \frac{5}{2}x - 1$$

10. [2 marks: 1, 1]

Find the gradient of the lines passing through the following pairs of points:

a)

$(3, -2)$ and $(-1, 6)$
 x_1, y_1, x_2, y_2

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{6 - (-2)}{-1 - 3}$$

$$= \frac{8}{-4}$$

$$= -2$$

b)

$(-4, -7)$ and $(1, -1)$
 x_1, y_1, x_2, y_2

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1 - (-7)}{1 - (-4)}$$

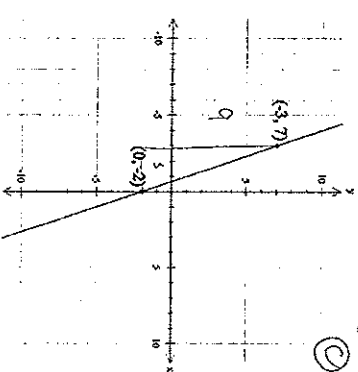
$$= \frac{6}{5}$$

(C)

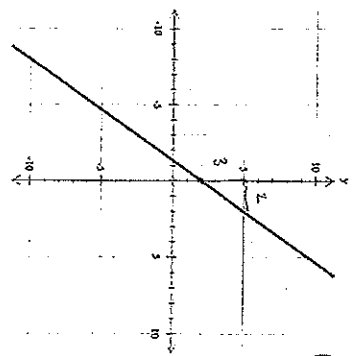
11. [4 marks: 2, 2]

Write the equation for each of the following lines:

a)



b)



(A)

$$y = -3x - 2$$

$$y = \frac{3}{2}x + 2$$

12. [7 marks: 1, 2, 2, 2]

Find the equation for the straight line with:

$$y = -4x$$

a) Gradient = -4 and passing through the origin

$$y = mx + c$$

$$-4 = 3(-2) + c$$

$$-4 = -6 + c$$

$$2 = c$$

$$\therefore y = 3x + 2$$

b) Gradient = 3 and passing through the point (-2, -4)

$$y = mx + c$$

$$0 = -\frac{1}{2}(-4) + c$$

$$0 = 2 + c$$

$$-2 = c$$

$$\therefore y = -\frac{1}{2}x - 2$$

c) Gradient = $\frac{3}{2}$ and passing through the point (-2, 5)

$$y = mx + c$$

$$5 = \frac{3}{2}(-2) + c$$

$$5 = -3 + c$$

$$8 = c$$

$$\therefore y = \frac{3}{2}x + 8$$

13. [6 marks: 2, 2, 2]

A hot air balloon descends (goes down) at a constant rate so that its height is 600 m after 30 seconds and after 60 seconds its altitude is 480 m.

a) Find the equation of the line that relates the height (h) of the balloon in metres and the time (t) in seconds.

$$(30, 600) \quad (60, 480)$$

$$m = \frac{600 - 480}{30 - 60}$$

$$= -4$$

$$y = -4x + c$$

$$600 = -4(30) + c$$

$$600 = -120 + c$$

$$720 = c$$

$$\therefore h = -4t + 720$$

b) Use the rule to find the height of the balloon after 2 minutes (120 sec).

$$h = -4(120) + 720$$

$$= -480 + 720$$

$$= 240 \text{ m}$$

c) Use the rule to show how long it takes for the balloon to return to the ground.

$$h = -4t + 720$$

$$\text{At } h = 0$$

$$0 = -4t + 720$$

$$-720 = -4t$$

$$180 = t$$

$$\therefore 180 \text{ sec}$$

$$3 \text{ minutes}$$